LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 1 Solutions

Name: _____

Campus ID: _____

Total Marks: 10
Time Duration: 15 minutes

Question 1 (2 marks)

Decide if each set is a subspace of $\mathbb{R}^3$. Give a one-line reason each.

(a) All vectors with $z = 0$.

(b) All vectors with $x + y + z = 1$.

Solution: (a) *Yes.* The xy-plane through the origin: contains $\mathbf{0}$ and is closed under addition and scalar multiplication (dimension 2).

(b) *No.* Does not contain $\mathbf{0}$ since $0 + 0 + 0 \neq 1$; also fails scalar-closure.

Question 2 (2 marks)

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ with $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$. Consider their inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \alpha$. Which of the following must always be true?

a. $\|\mathbf{u} - \mathbf{v}\|^2 = 2(1 - \alpha)$
b. $\|\mathbf{u} + \mathbf{v}\|^2 = 2(1 - \alpha)$
c. If $\alpha = 0$, then $\mathbf{u}$ and $\mathbf{v}$ are parallel.
d. If $\alpha = -1$, then $\mathbf{u} = \mathbf{v}$.

Solution: By expanding norms:

$$\|\mathbf{u} - \mathbf{v}\|^2 = \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle = 1 + 1 - 2\alpha = 2(1 - \alpha).$$

Other statements are false: $\|\mathbf{u} + \mathbf{v}\|^2 = 2(1 + \alpha)$, $\alpha = 0$ means orthogonal (not parallel), and $\alpha = -1$ means $\mathbf{u} = -\mathbf{v}$.

Question 3 (6 marks)

Consider two vectors in $\mathbb{R}^3$

$$\mathbf{u}_2 = (1, 0, 1), \quad \mathbf{u}_1 = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0 \right).$$

(a) [1 mark] Describe $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ geometrically in $\mathbb{R}^3$.

(b) [5 marks] Apply the Gram–Schmidt process to the set $\{\mathbf{u}_1, \mathbf{u}_2\}$ to produce an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.
Hint: If one of $\mathbf{u}_1$ or $\mathbf{u}_2$ already has unit norm, choose it as $\mathbf{v}_1$.

Solution:

1. The span of these two vectors is
$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2\} = \{\alpha\mathbf{u}_1 + \beta\mathbf{u}_2 : \alpha, \beta \in \mathbb{R}\}.$$
Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are not scalar multiples of each other, they are linearly independent. **Therefore, their span is a 2-dimensional subspace of $\mathbb{R}^3$, i.e., a plane through the origin.**
Moreover, the plane spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$ has normal vector
$$\mathbf{n} = \mathbf{u}_1 \times \mathbf{u}_2 = \left(\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}, -\tfrac{1}{\sqrt{2}}\right).$$
Hence, the equation of the plane is
$$x - y - z = 0.$$
$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2\} = \{(x, y, z) \in \mathbb{R}^3 : x - y - z = 0\}.$$

2.
$$\|\mathbf{u}_1\|^2 = \left(\tfrac{1}{\sqrt{2}}\right)^2 + \left(\tfrac{1}{\sqrt{2}}\right)^2 + 0^2 = \tfrac{1}{2} + \tfrac{1}{2} = 1.$$
So $\mathbf{u}_1$ is unit-normalized. Now Orthogonalize $\mathbf{u}_2$ with respect to $\mathbf{u}_1$.

$$\mathbf{v}_2 = \mathbf{u}_2 - \langle \mathbf{u}_2, \mathbf{u}_1 \rangle \mathbf{u}_1.$$
$$\langle \mathbf{u}_2, \mathbf{u}_1 \rangle = (1, 0, 1) \cdot \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) = \tfrac{1}{\sqrt{2}}.$$
So,
$$\mathbf{v}_2 = (1, 0, 1) - \tfrac{1}{\sqrt{2}} \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) = (1, 0, 1) - \left(\tfrac{1}{2}, \tfrac{1}{2}, 0\right) = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right).$$
Now normalize the result to obtain an orthonormal basis.

$$\|\mathbf{v}_2\| = \sqrt{\left(\tfrac{1}{2}\right)^2 + \left(-\tfrac{1}{2}\right)^2 + 1^2} = \sqrt{\tfrac{1}{4} + \tfrac{1}{4} + 1} = \sqrt{\tfrac{3}{2}}.$$
So,
$$\mathbf{u}_2^* = \frac{1}{\sqrt{3/2}} \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right) = \left(\tfrac{1}{\sqrt{6}}, -\tfrac{1}{\sqrt{6}}, \tfrac{2}{\sqrt{6}}\right).$$
The orthonormal set (using $\mathbf{u}_1$ first) is
$$\left\{\left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right), \left(\tfrac{1}{\sqrt{6}}, -\tfrac{1}{\sqrt{6}}, \tfrac{2}{\sqrt{6}}\right)\right\}.$$

Alternative solution (ignoring the hint, starting with $\mathbf{u}_2$):
First normalize $\mathbf{u}_2$:
$$\|\mathbf{u}_2\| = \sqrt{1 + 0 + 1} = \sqrt{2}, \qquad \mathbf{v}_1' = \frac{\mathbf{u}_2}{\sqrt{2}} = \left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right).$$
Now orthogonalize $\mathbf{u}_1$ against $\mathbf{v}_1'$:
$$\langle \mathbf{u}_1, \mathbf{v}_1' \rangle = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) \cdot \left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right) = \tfrac{1}{2}.$$
$$\mathbf{w}' = \mathbf{u}_1 - \tfrac{1}{2}\mathbf{v}_1' = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 0\right) - \tfrac{1}{2}\left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right) = \left(\tfrac{1}{2\sqrt{2}}, \tfrac{1}{\sqrt{2}}, -\tfrac{1}{2\sqrt{2}}\right).$$
Simplify:
$$\mathbf{w}' = \tfrac{1}{2\sqrt{2}}(1, 2, -1).$$
Its norm:
$$\|\mathbf{w}'\| = \sqrt{\tfrac{1}{8} + \tfrac{1}{2} + \tfrac{1}{8}} = \sqrt{\tfrac{3}{4}} = \tfrac{\sqrt{3}}{2}.$$
Normalize:
$$\mathbf{v}_2' = \frac{\mathbf{w}'}{\|\mathbf{w}'\|} = \frac{\frac{1}{2\sqrt{2}}(1, 2, -1)}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{6}}(1, 2, -1).$$
So the orthonormal set (using $\mathbf{u}_2$ first) is
$$\left\{\left(\tfrac{1}{\sqrt{2}}, 0, \tfrac{1}{\sqrt{2}}\right), \tfrac{1}{\sqrt{6}}(1, 2, -1)\right\}.$$